

Arsenic Demo Sites: Where Are They Now? A Retrospective Cost and Performance

Analysis of EPA's 50 Full-Scale Treatment Pilots

Presenter/Author: Natchaya Luangphairin (natchayal@usf.edu)
Co-Author and PI: Manish Adhikari, Katherine Alfredo, PhD, PE
University of South Florida, Department of Civil and Environmental Engineering, Tampa, FL

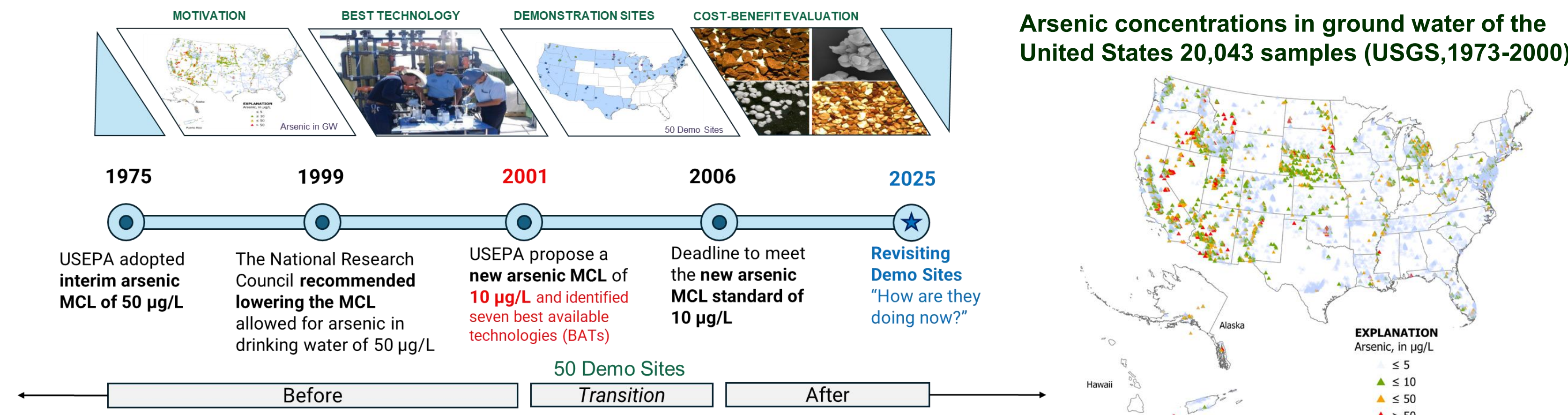
Abstract

In 2001, the U.S. EPA reduced the arsenic MCL from 50 µg/L to 10 µg/L, and expert projections indicated that more than 20% of public water systems—particularly small groundwater systems—would be unable to meet the new standard. To support utilities, EPA implemented 50 demonstration projects to evaluate multiple treatment technologies under real operating conditions. This study revisits those systems to assess long-term cost, performance, and sustainability.

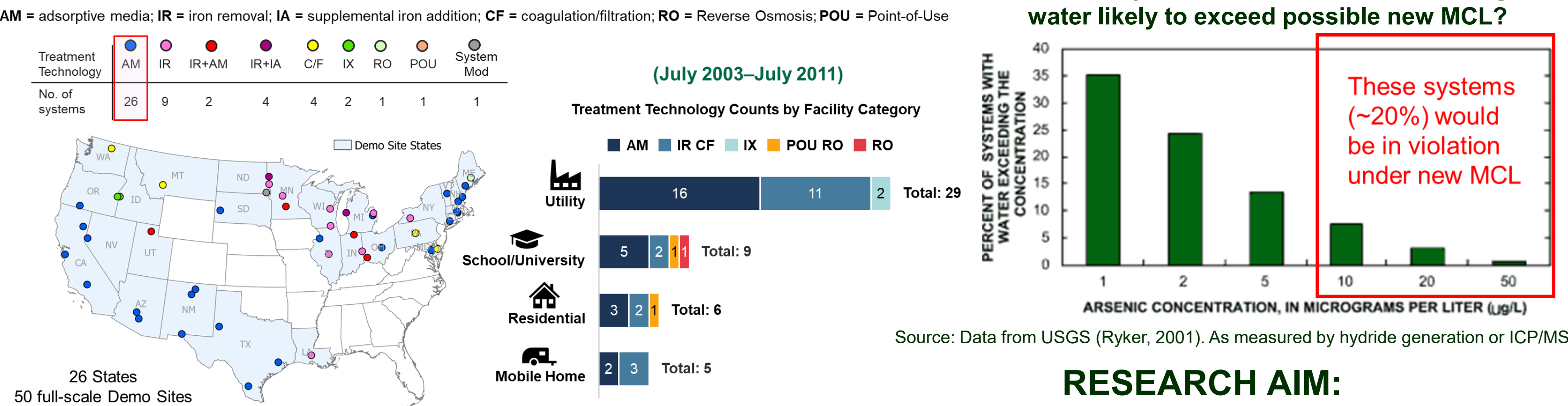
This research uses EPA's cost reports and water quality datasets to compare removal efficiency, compliance reliability, and operational behavior across technologies, with a specific focus on adsorptive media. The results identify key design and operational drivers linked to cost-effectiveness and sustained regulatory compliance

Background & Problem Identification

The Hidden Threat: Arsenic in Drinking Water Timeline and Regulation



EPA's Response: 50 Full-Scale Demonstration

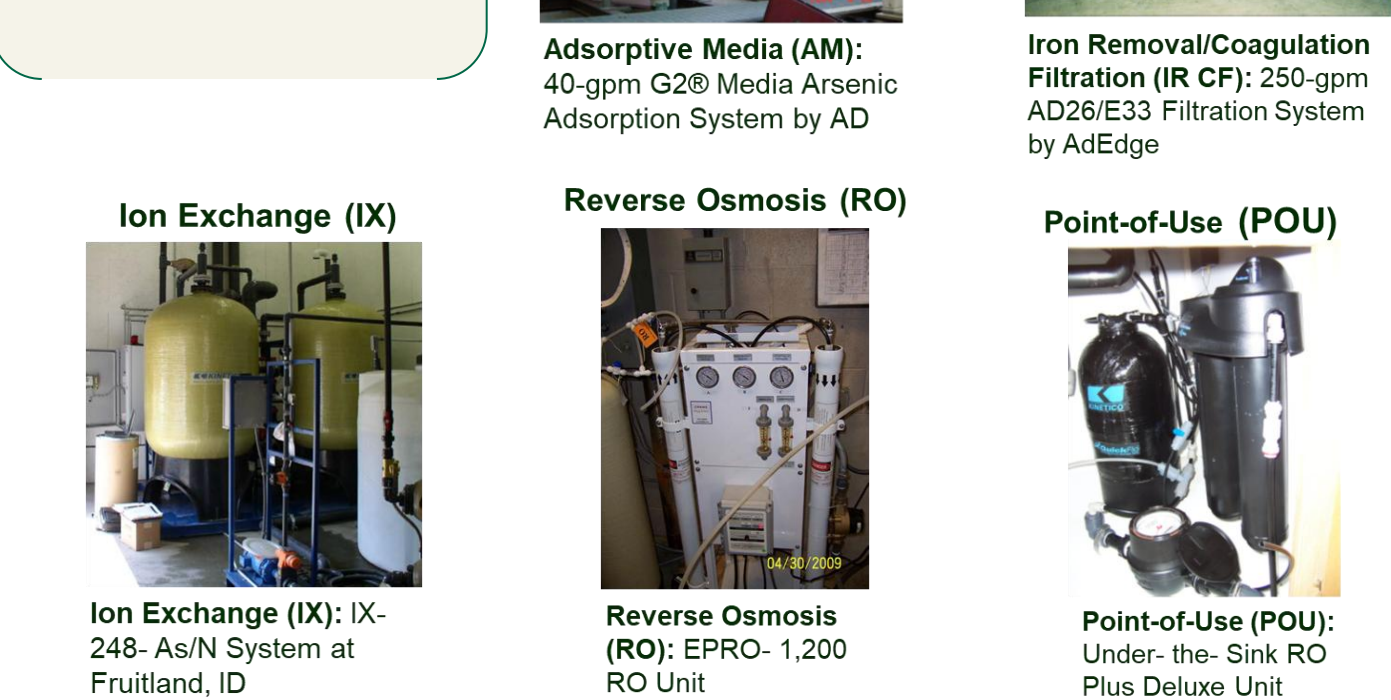


RESEARCH AIM:
To evaluate which arsenic treatment technology delivers the best balance of performance and cost.

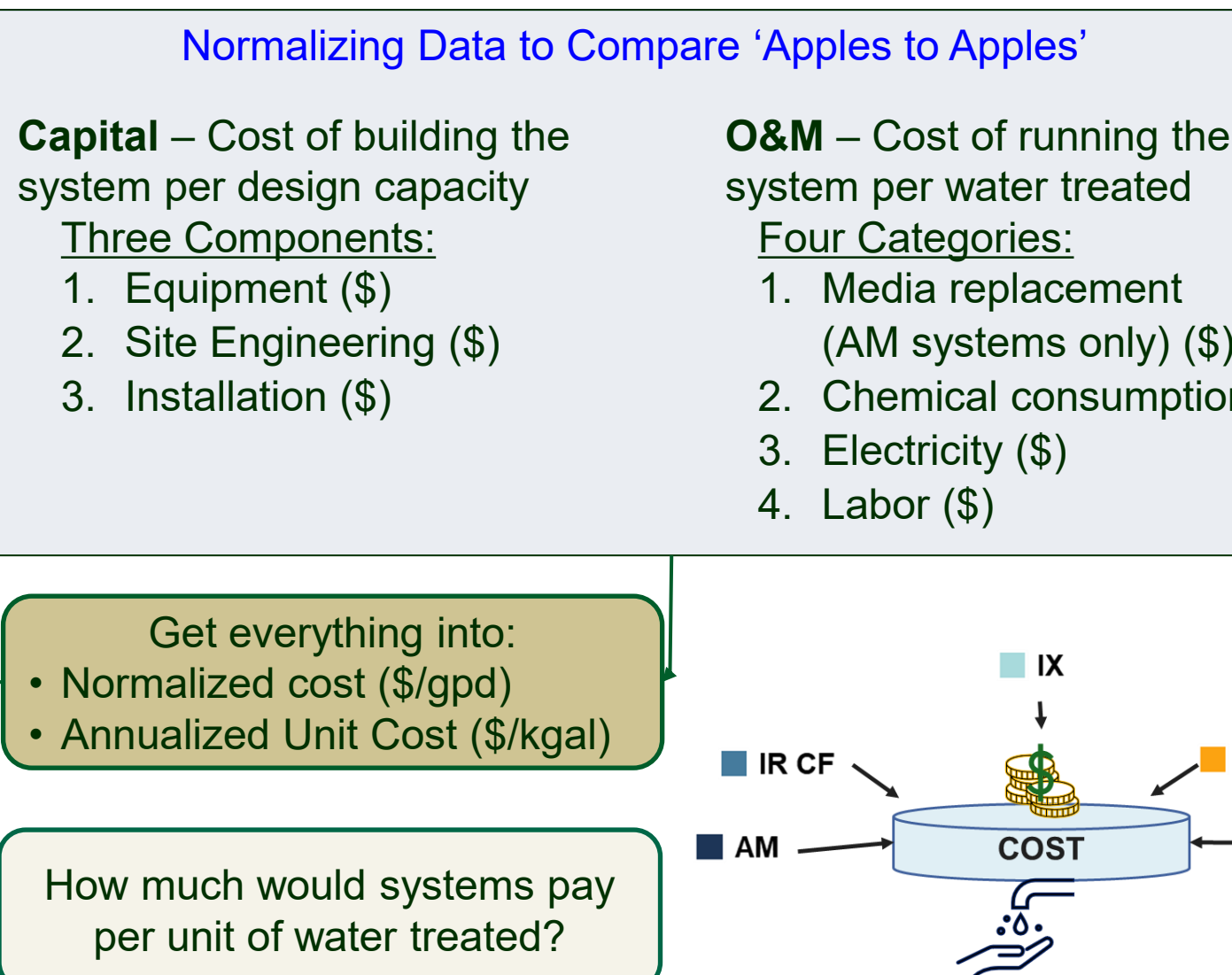
RESEARCH AIM:
To examine whether small systems—originally projected to be most at risk of non-compliance—achieved the 10 µg/L arsenic MCL after adopting treatment technologies.

Research Approach

Technologies on Trial



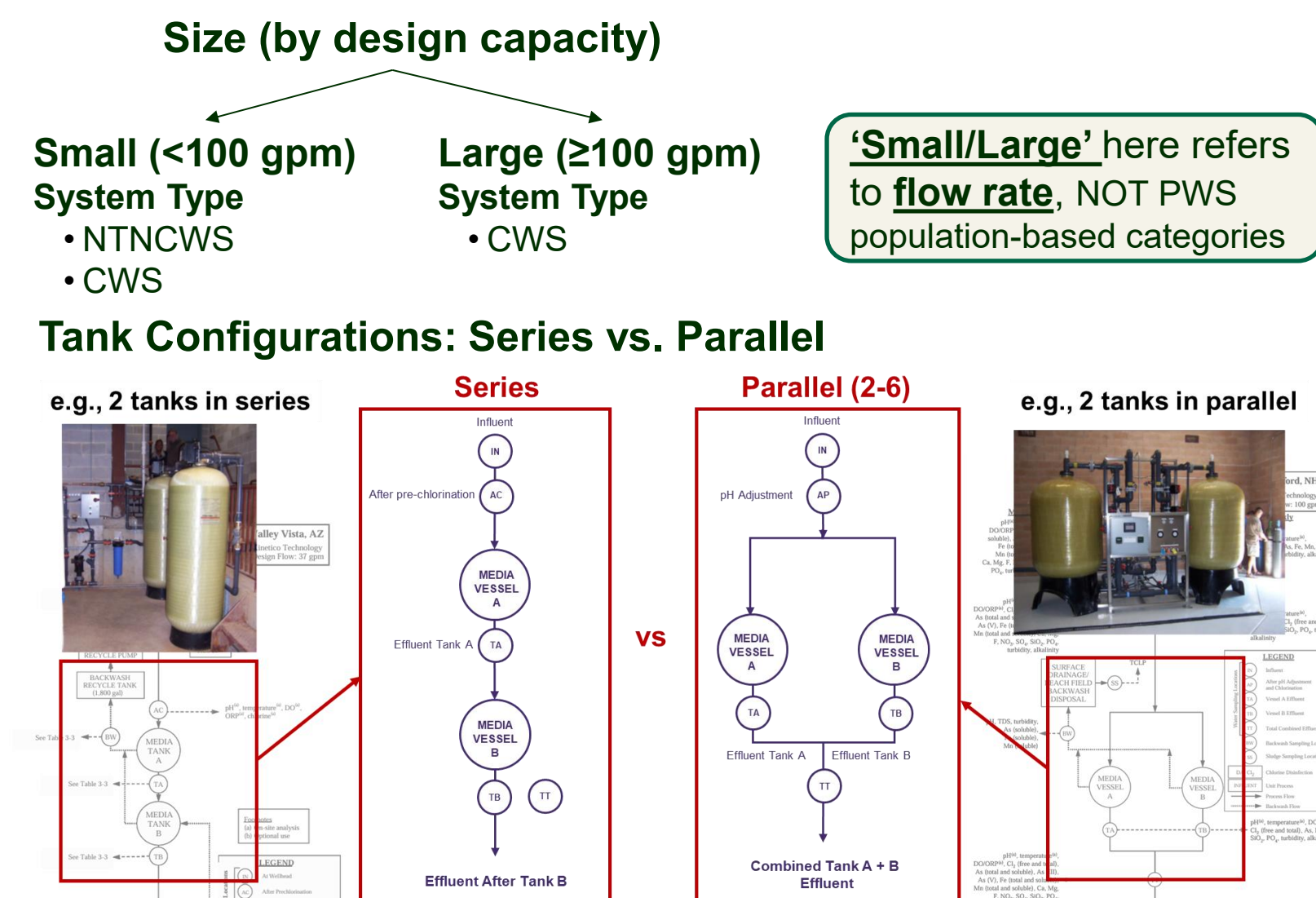
Cost Analysis for All Treatment Technologies



Nine Different Adsorptive Media Used (28 AM Systems)

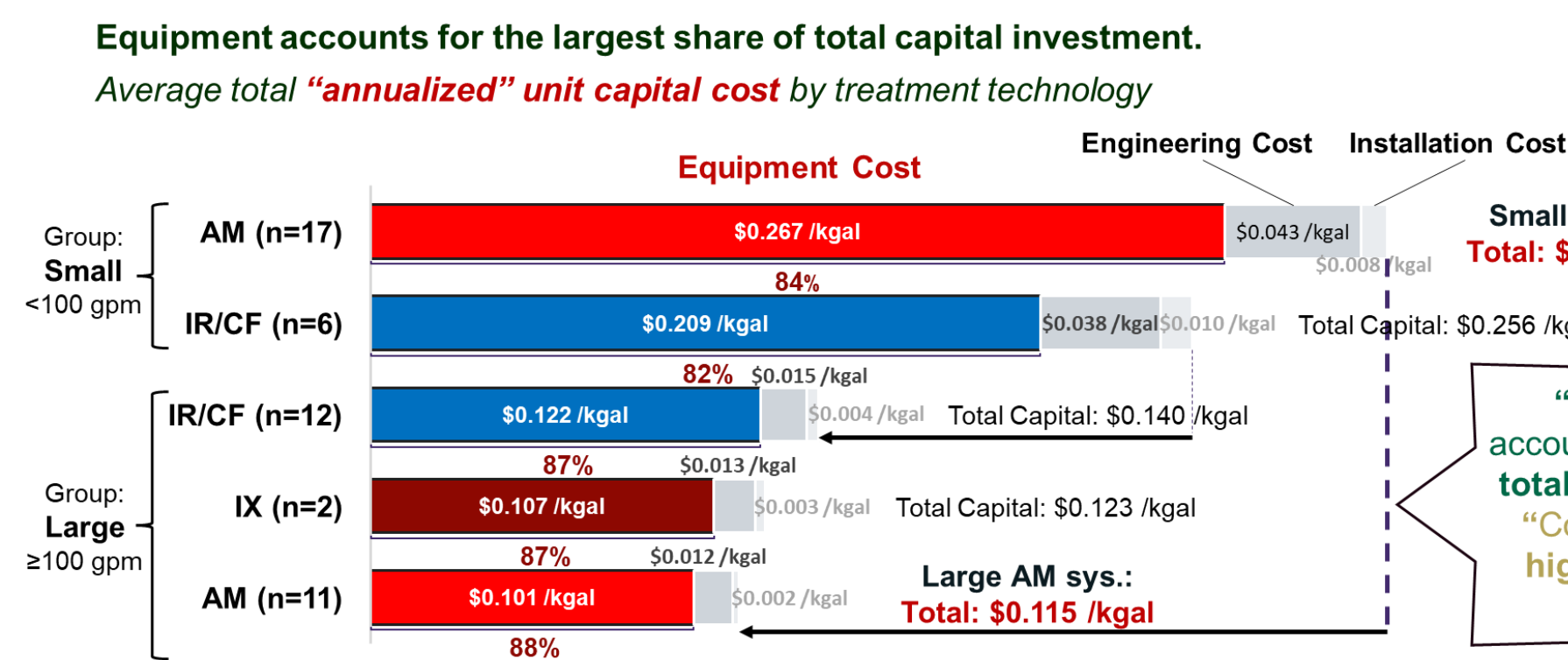


Performance Analysis for Adsorptive Media Systems

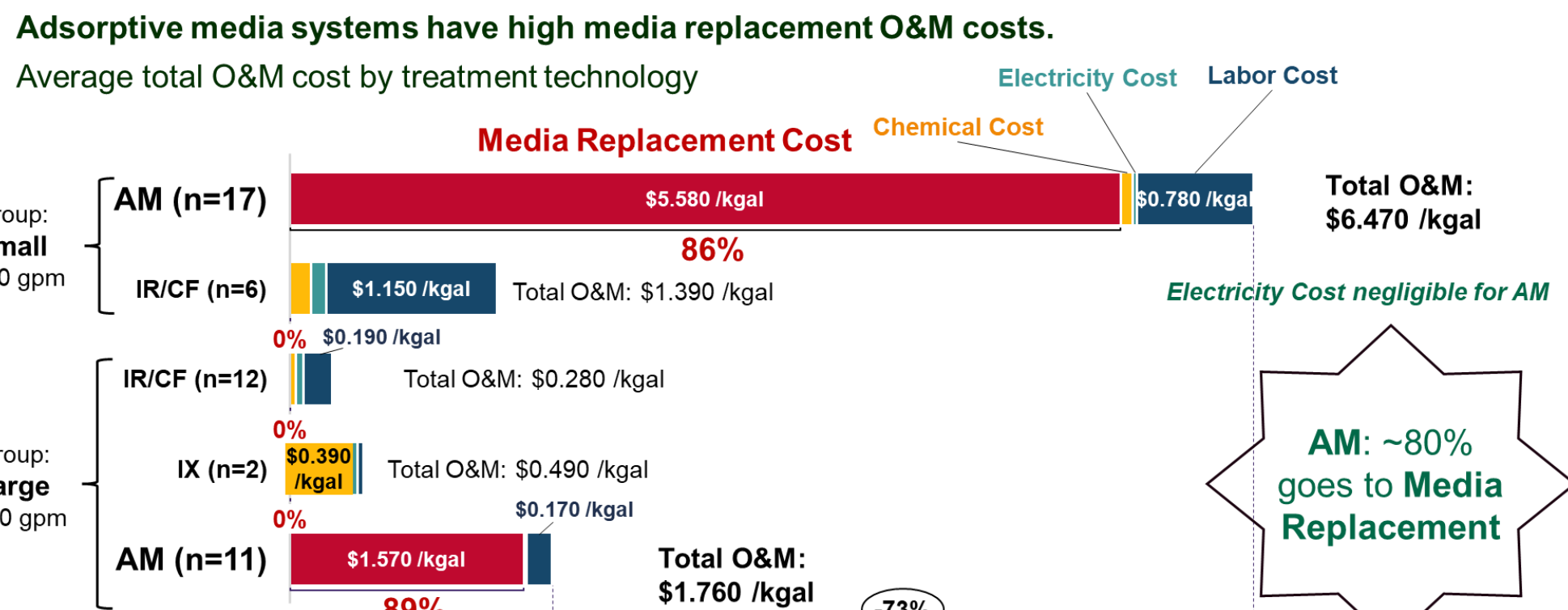


Results & Discussion

Which Component Drives Capital Costs?

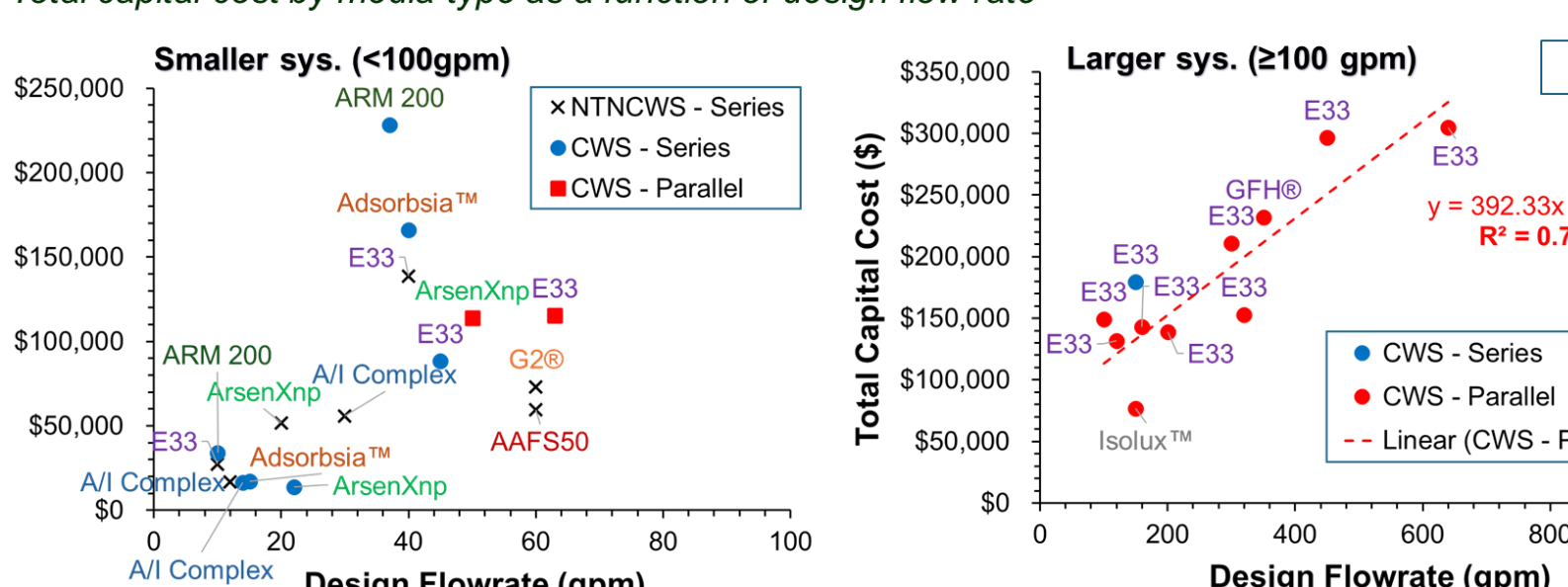


Which Component Drives O&M Costs?

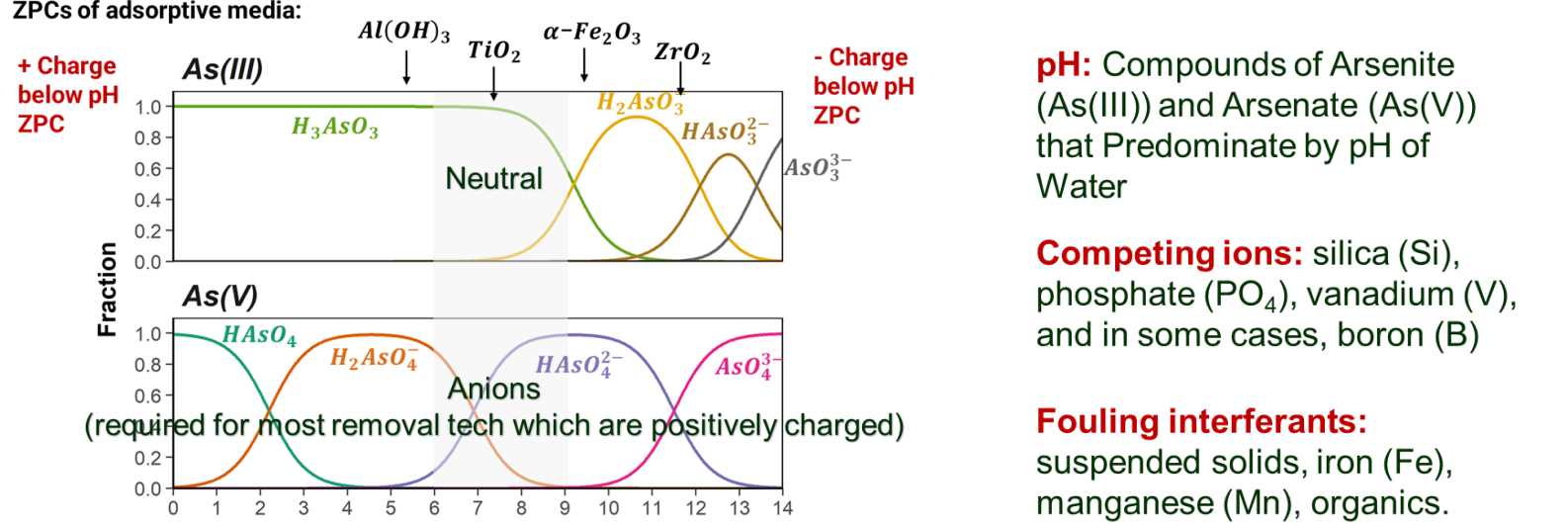


[Case Studies: Adsorptive Media (AM) Cost and Performance]

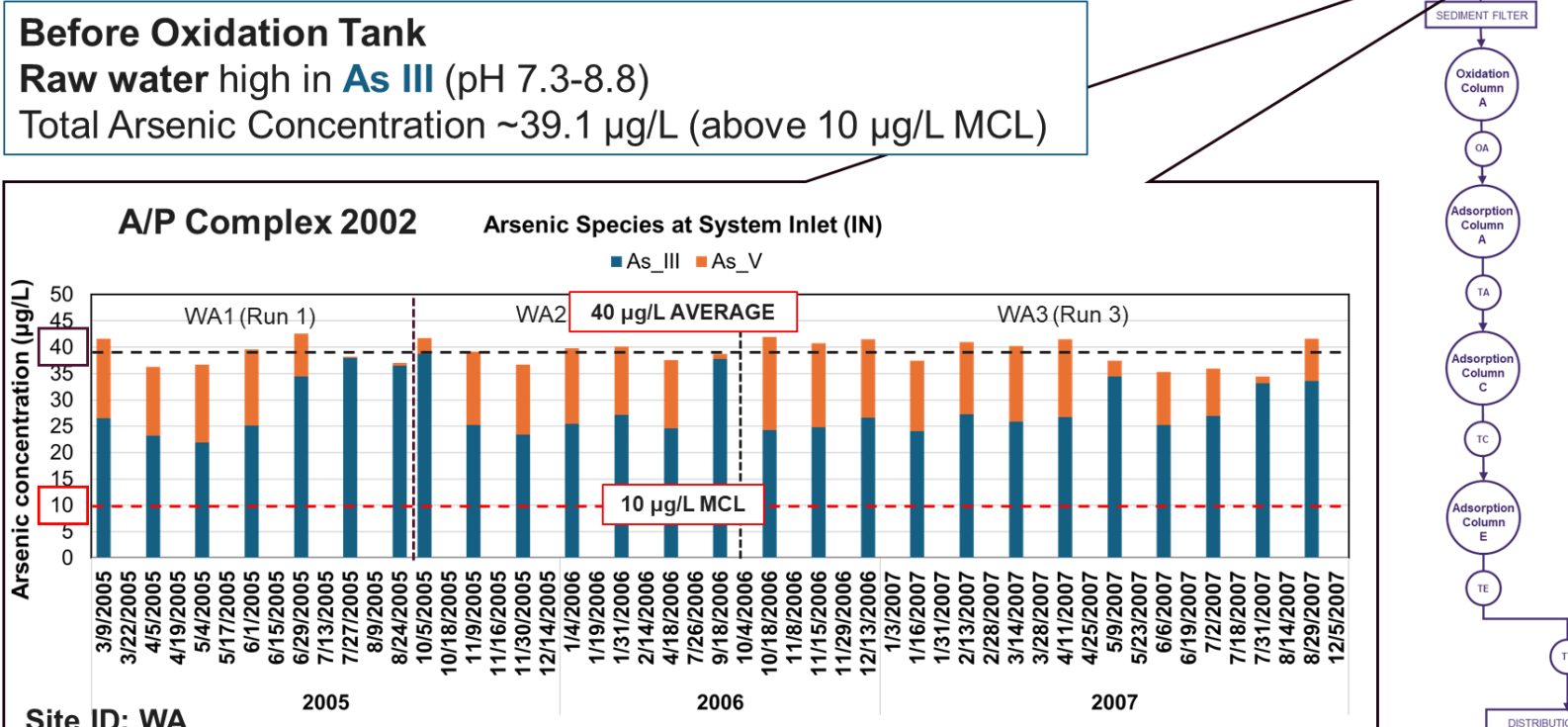
AM: Iron-based (E33) media popular among larger Adsorptive Media systems.



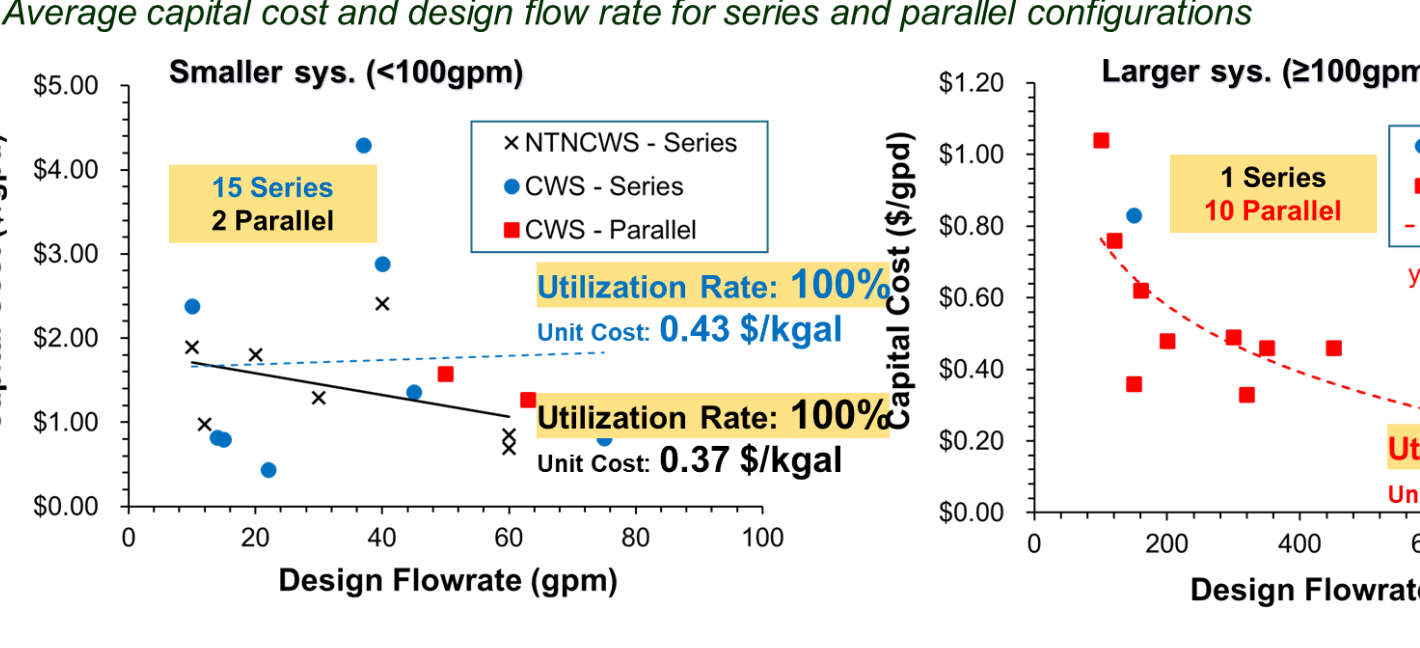
[Water Quality: Diverse Sites, Diverse Challenges]



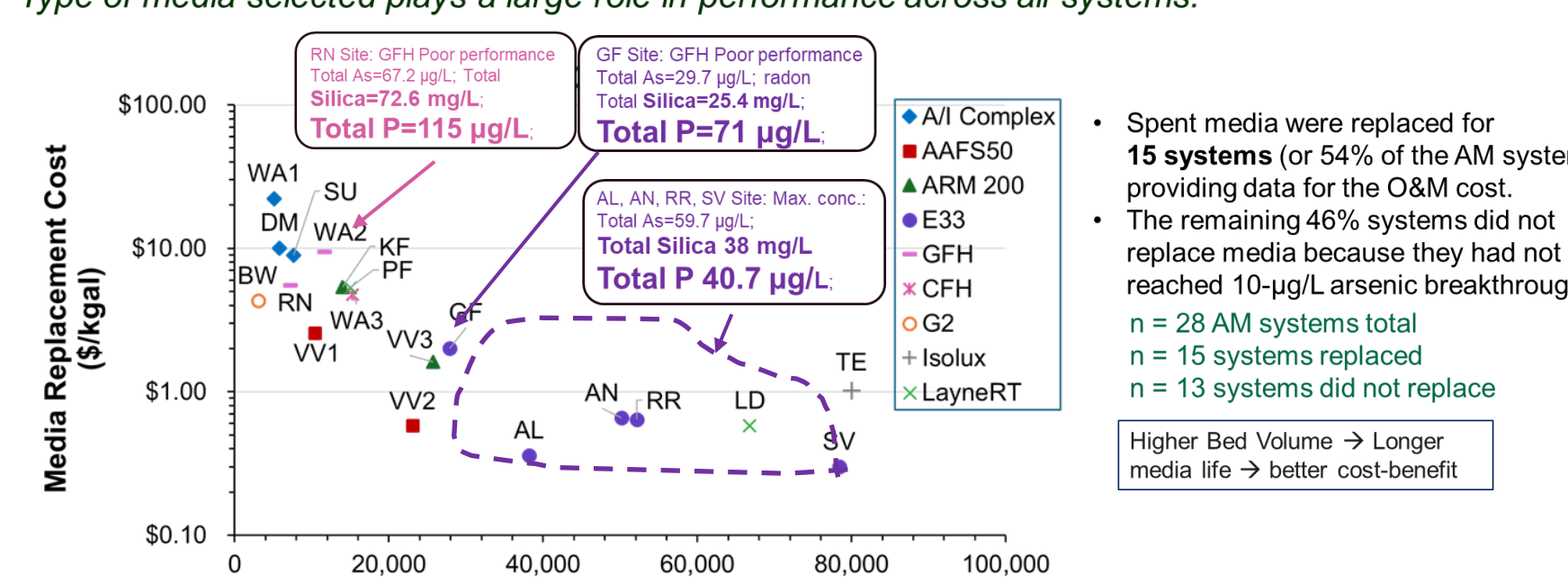
[Lessons Learned from the Field]



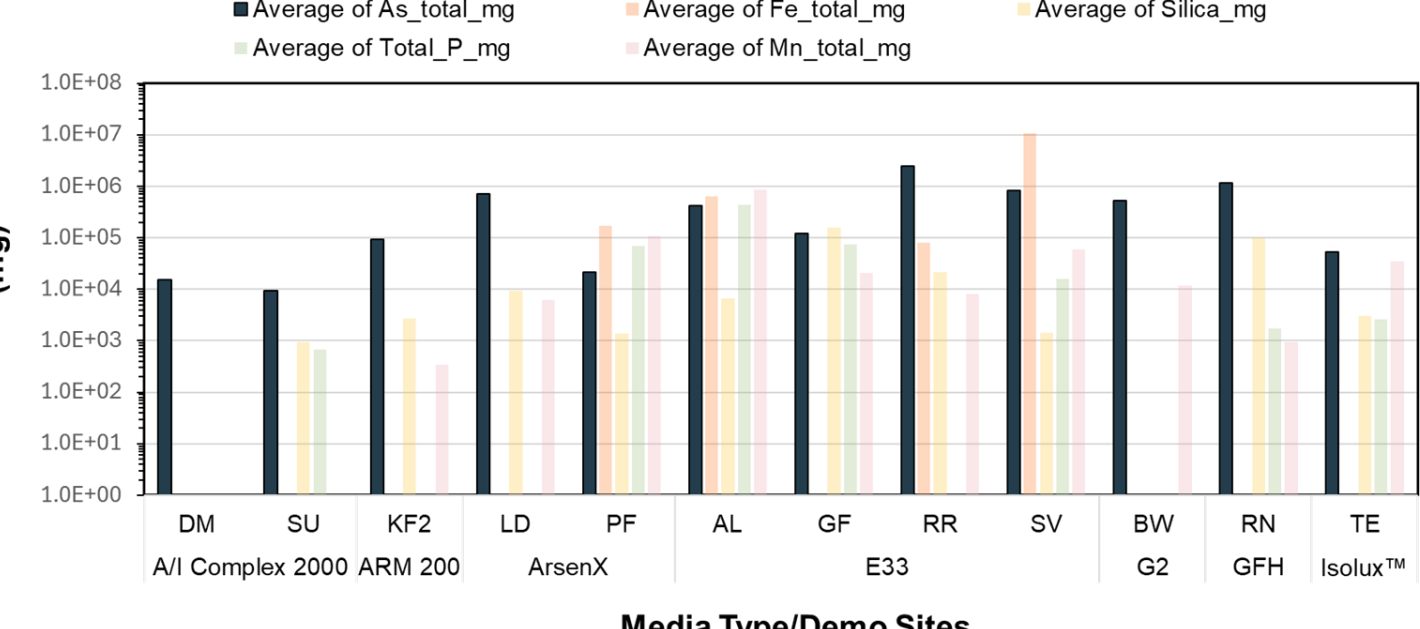
Larger CWS typically have higher hydraulic utilization rate reducing total unit cost.



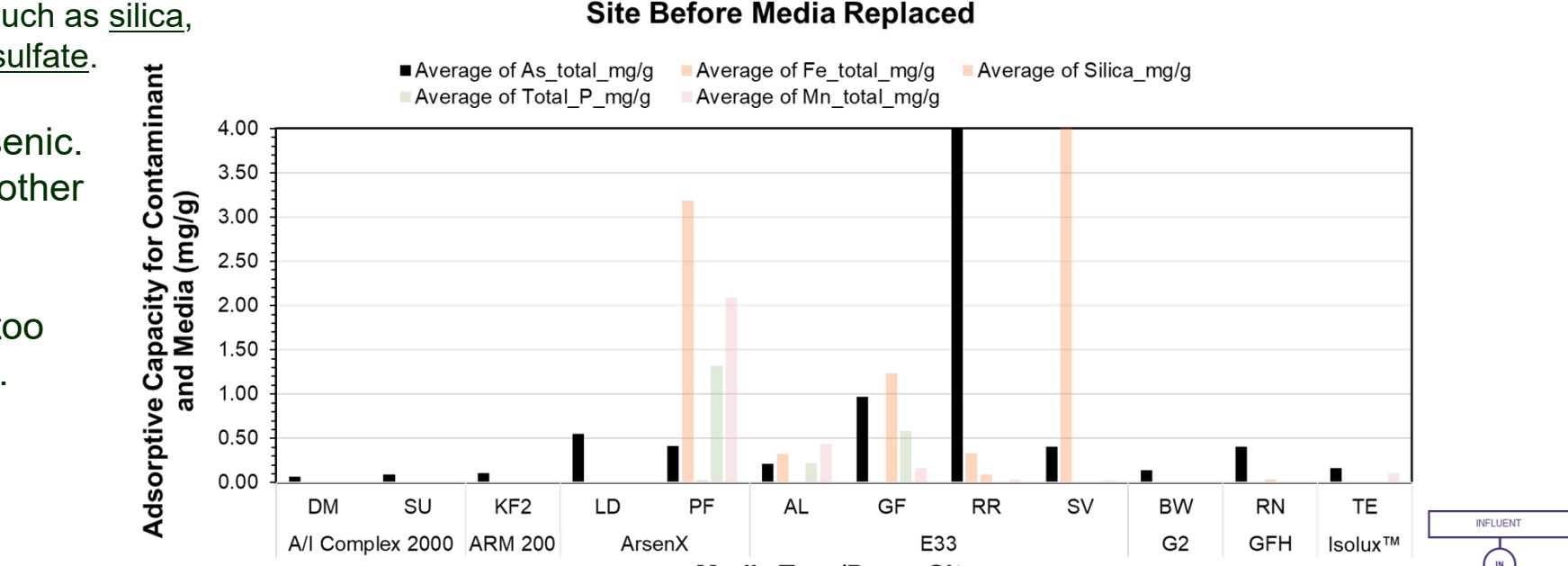
Different media types have different adsorptive capacity and selectivity impacting run length.



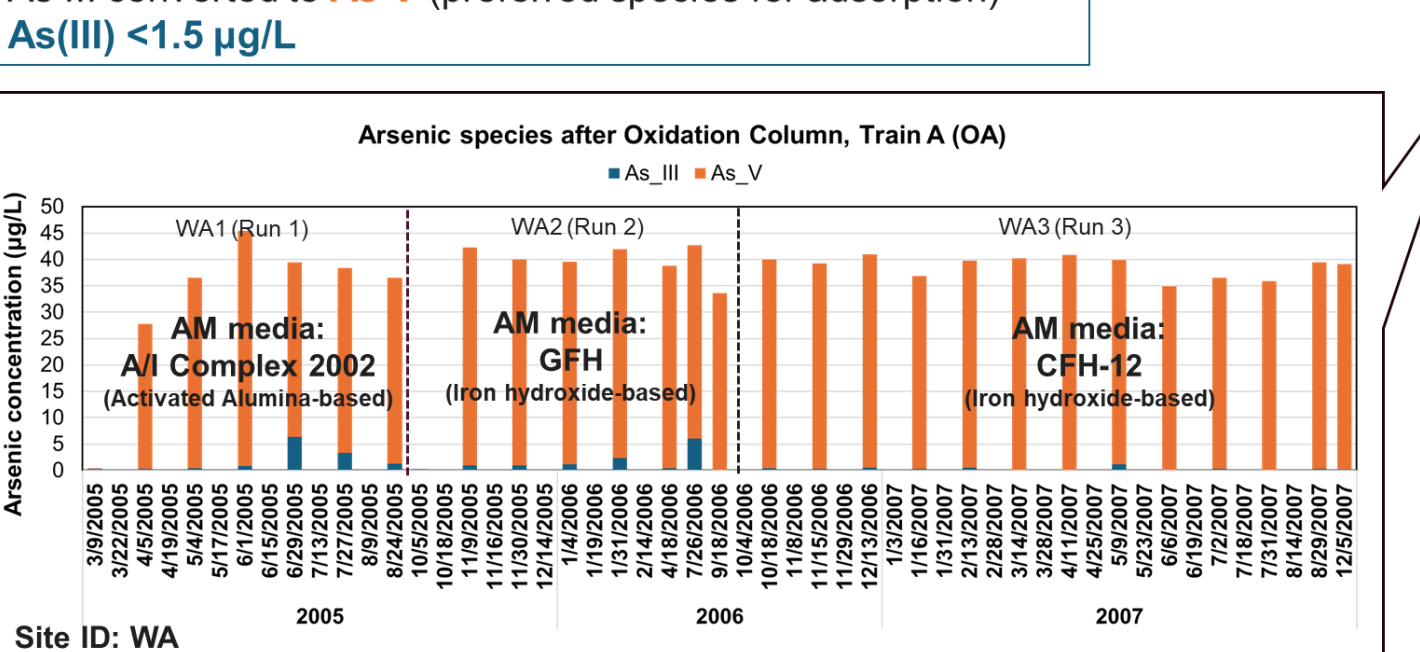
Total Contaminant Removed (mg) Before Media Replaced



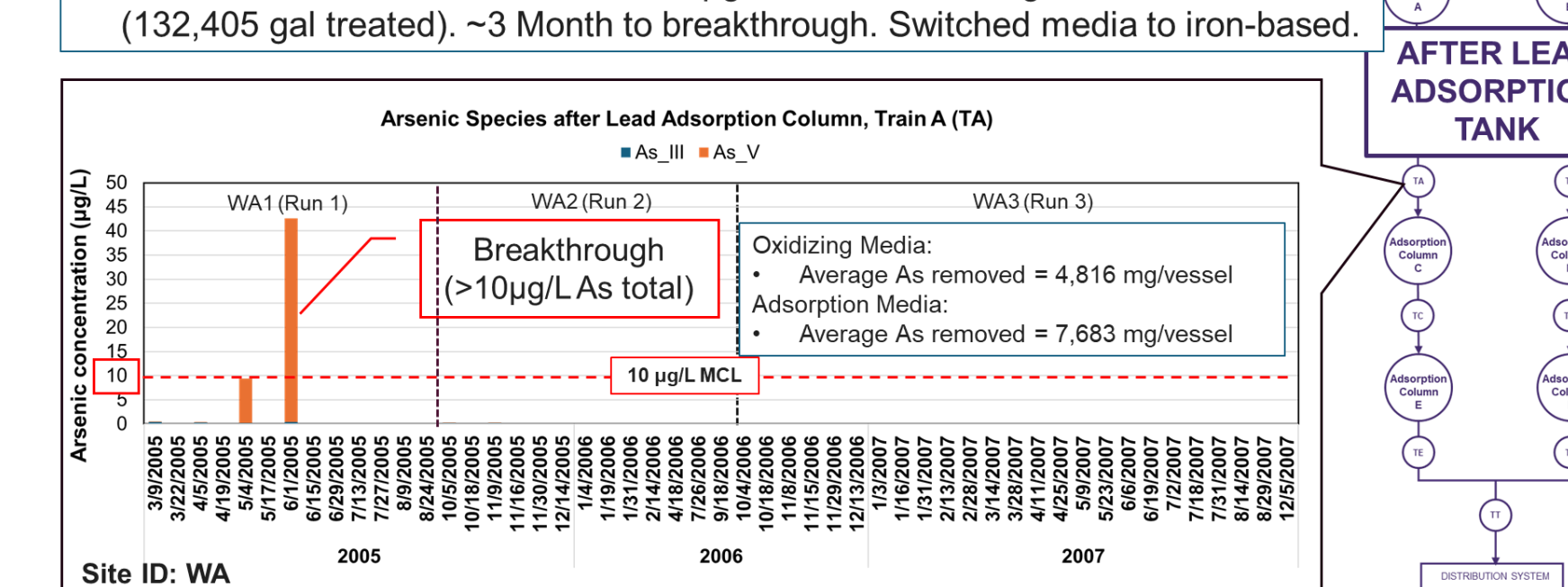
Media Run Length (BV)



After Oxidation Tank



After Lead Adsorption Tank



Conclusions & Ongoing Work

Putting it all together:

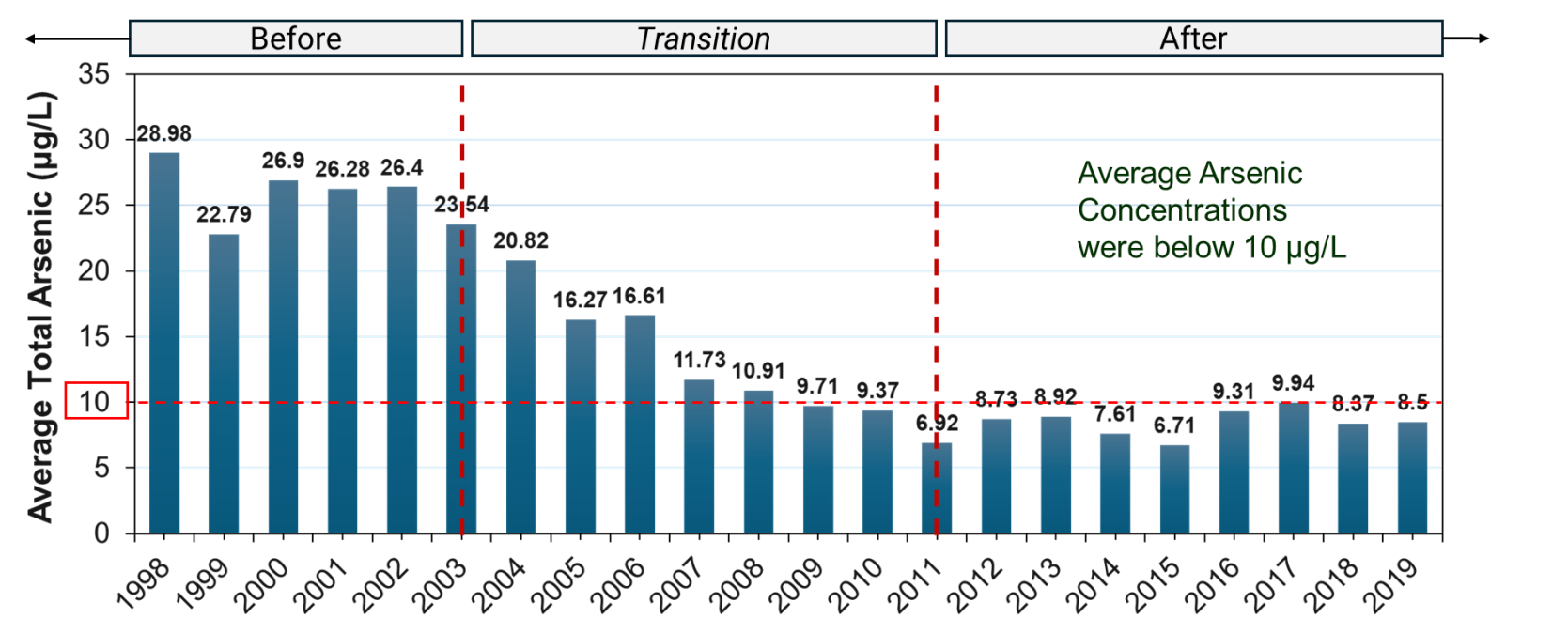
Treatment Technology	No. of Systems	Design Flow Rate (gpm)	Total Capital Cost (\$)	Total Capital Unit Cost (\$/kgal)	Total O&M Costs (\$/kgal)
AM (<100 gpm)	17	10–75	14,000–228,309	0.41	6.47
IR/CF (<100 gpm)	6	20–96	55,423–161,560	0.38	1.39
AM (≥100 gpm)	11	100–640	74,840–305,000	0.14	1.76
IR/CF (≥100 gpm)	12	140–770	216,876–427,407	0.19	0.28
IX (≥100 gpm)	2	250–640	286,388–395,434	0.17	0.49

Configuration	Main Benefit	Ideal For	Cost Impact	Trade-Off
Lead-Lag (Series)	Ensures compliance — lag vessel captures breakthrough	Small to mid systems, drinking water, high contaminant risk	Slightly higher capital and operating cost (two vessels in series, higher pressure drop)	Lower flow capacity but maximized media use and reliability
Parallel	Provides higher flow and operational flexibility	Large systems or moderate-risk applications	Lower cost per flow unit — simpler piping and lower head loss	No redundancy; both vessels may break through together

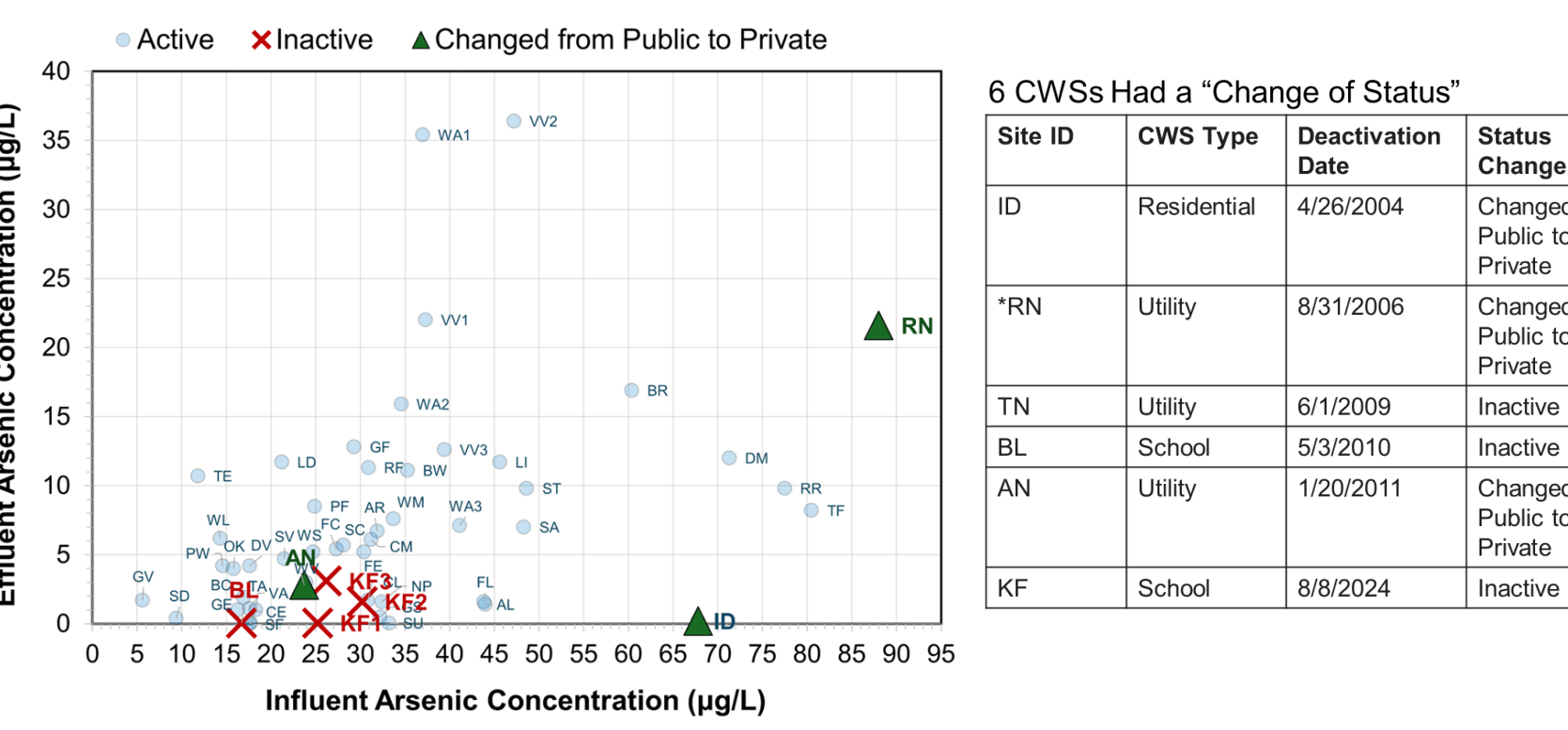
- By 2006, nearly all systems achieved compliance with the 10 µg/L MCL.
- Iron-based AM delivered the best balance between performance and cost.
- Adsorption capacity of media can be impacted by high levels of competing ions such as silica, phosphate, and sulfate.
- Top contributors of high system costs are pH control system, size of vessels, and more advanced system control.
- Lead/Lag configuration allows for better utilization of media for Arsenic removal.
- Higher utilization → better cost efficiency...Lower utilization → higher unit cost.

The results from both studies highlight that iron-based AM systems consistently emerge as the most cost-effective option for small systems to achieve and maintain arsenic compliance.

Overall Compliance Tracking: Average Total Arsenic Concentrations of All 50 Demo Sites Over Time



Technology Longevity: Who Stayed, Who Switched?



E33 iron-based media showed the lowest cost and longest media life. Equipment and media replacement costs dominated total expenditures. All systems began above the 10 µg/L MCL, but by 2006 nearly all achieved compliance. Arsenic removal was strongest in lower-pH water with fewer competing ions.

Ongoing Research:

- Conduct follow-up surveys with vendors and utilities to validate operational and performance data.
- Expand detailed analysis of performance beyond adsorptive media (AM) to include IX, CF, and RO systems.
- Further investigate whether any systems transitioned to different treatment technologies over time and what factors drove those changes.
- Perform advanced statistical modeling to explore and quantify variable impacts on arsenic removal:
 - Multilevel / Hierarchical models — capture variation across sites and technologies.
 - Survival analysis (time-to-event) — estimate and compare media lifespans.
 - Variable importance analysis — quantify the weight of impact of pH, phosphate, silica, and other factors on removal efficiency.

For more information, comments, or questions on this research, please contact:

Natchaya Luangphairin
natchayal@usf.edu
Katherine Alfredo, PhD, PE
kalfredo@usf.edu

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EPA reports: <https://www.epa.gov/water-research/arsenic-treatment-technology-demonstrations>